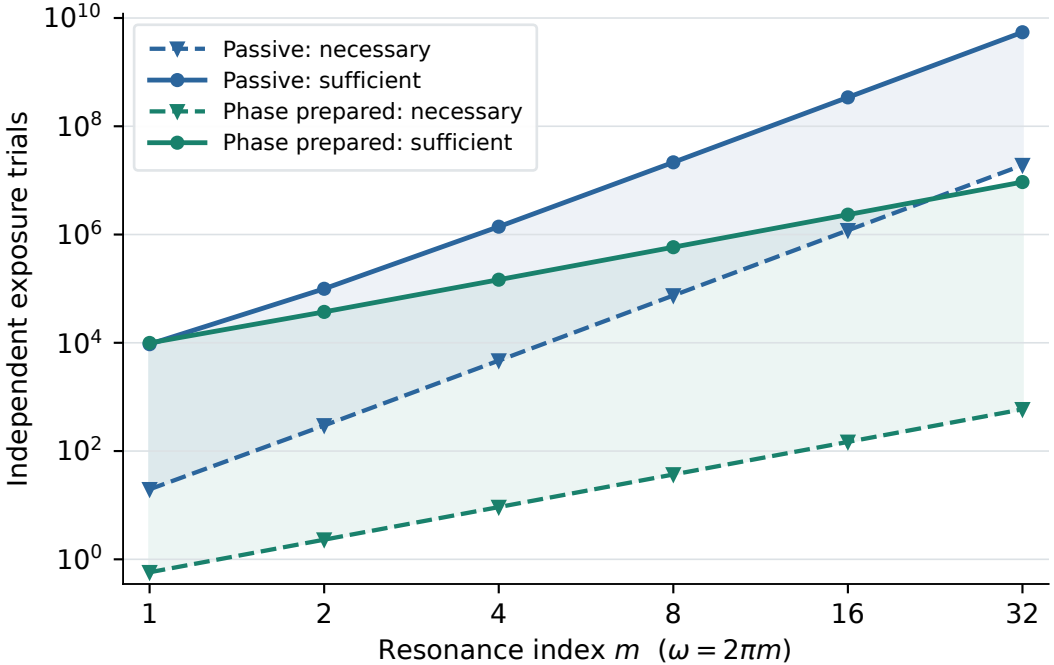
Detector access changes the exposure count

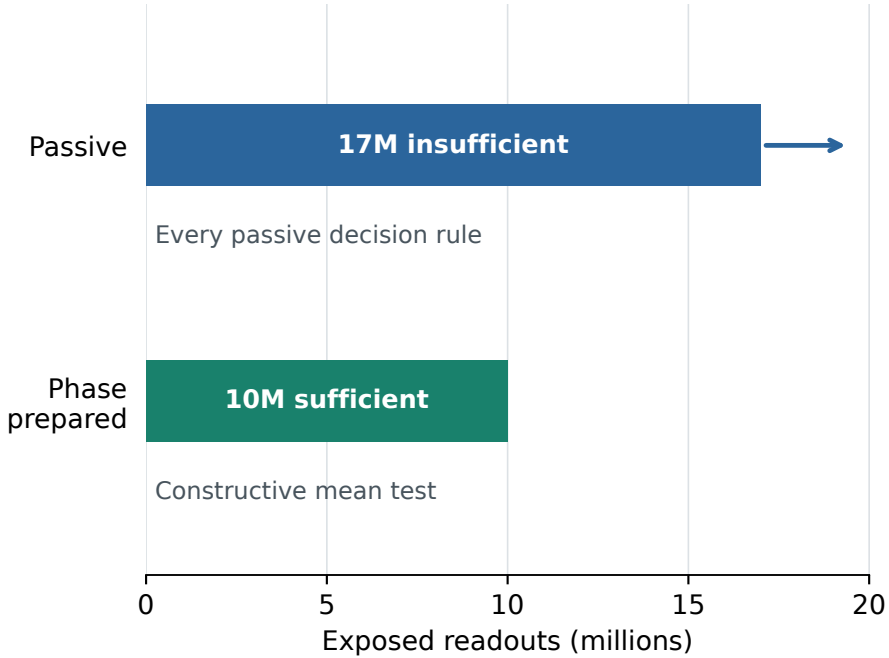
Known alternatives ω and 2ω; $\gamma = \tau = 1$, Gaussian noise $\nu = 1/2$, target binary error ≤ 0.1.

Analytical count brackets



Shading bounds the required count; it is not a confidence band.

Exact comparison at $m = 32$



This finite separation uses exact rational inequalities.

Fixed read noise. Each trial requires an independent preparation; preparation time and energy are not counted.

Under the separately assumed sensor law $\nu^2 = \kappa/E$, both optimized read-energy orders are $\Theta(\omega^2)$.